# MNIST-LOSS AND ACCURACY

```
model.compile(
    optimizer="adam",
    loss="sparse_categorical_crossentropy",
    metrics=["accuracy"]
)

history = model.fit(
    x_train,
    y_train,
    epochs=10,
    batch_size=128,
    validation_split=0.1
)

plt.figure(figsize=(8, 5))

plt.plot(history.history["accuracy"], label="Training Accuracy")
plt.plot(history.history["val_accuracy"], label="Validation Accuracy")

plt.title("CNN Training and Validation Accuracy")
plt.xlabel("Epoch")
plt.ylabel("Accuracy")
plt.legend()
plt.grid(True)
plt.show()

plt.figure(figsize=(8, 5))

plt.plot(history.history["loss"], label="Training Loss")
plt.plot(history.history["val_loss"], label="Validation Loss")

plt.title("CNN Training and Validation Loss")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.legend()
plt.grid(True)
plt.show()

test_loss, test_accuracy = model.evaluate(
    x_test,
    y_test,
    verbose=2
)

print("Test Loss:", test_loss)
print("Test Accuracy:", test_accuracy)
print("Test Accuracy (%):", test_accuracy * 100)
```